Type MS · Describe what red cells and plasma carry, and from where to where

Answer 1 of 6 · Question 6 part b · 5 marks

2024 JUN · Paper 01 · find part b in this mark scheme extract

| | | | |
|---|---|---|---|
| 6 (a) (i) | 3 x 3 x 6;<br>54 ($cm^2$) | full marks for correct final answer ecf | (2) |
| (ii) | 3 x 3 x 3;<br>27 ($cm^3$) | full marks for correct final answer ecf | (2) |
| (iii) | 54:27;<br>2:1 | ecf for wrong values | (2) |
| (b) | Any five from<br>• red blood cells contain haemoglobin;<br>• to transport oxygen/binds to oxygen;<br>• blood plasma;<br>• contains dissolved substances;<br>• named substance e.g. nutrients/hormones/$CO_2$/heat;<br>• transported through blood vessels/named blood vessel; | | (5) |
| (c) | • diffusion;<br>• (nutrients) higher in concentration outside the cell than inside the cell/down a concentration gradient/from high to low concentration; | | (2) |

**Total for Question 6 = 13 marks**

Type MS · Describe what red cells and plasma carry, and from where to where

Answer 2 of 6 · Question 1 part a · 1 mark

2020 NOV · Paper 01 (regional R) · find part a in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 1 | (a) | transport oxygen to body cells | | 1 |
| | (b) | • (X) plasma (1)<br>• (Y) platelets (1) | | 2 |
| | (i) | Evidence of division by 5(1)<br>3:1 | full marks for correct final answer | 2 |
| | (ii) | • white blood cells/lymphocytes increase in number (1)<br>• (white blood cells)used in defence against disease(1)<br>• produce antibodies/phagocytosis(1) | Ignore fight disease/pathogen<br>Allow kill/destroy pathogen/bacteria | 2 |

**Total for Question 1 = 7 marks**

Type MS · Describe what red cells and plasma carry, and from where to where

Answer 3 of 6 · Question 1 part a.i · 3 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| 1 (a) (i) | • transport oxygen;<br>• from lungs;<br>• to (body) tissues/cells; | Allow organs | 3 |
| (ii) | • haemoglobin to carry/combine with oxygen;<br>• no nucleus, so more haemoglobin;<br>• biconcave shape to increase surface area;<br>• small/flexible (cell) membrane to pass through capillaries;<br>• thin membrane to reduce diffusion distance; | | 4 |
| (b) (i) | vein; | | 1 |
| (ii) | • valve present/large lumen/thin (muscular) walls;<br>• only found in veins/not found in arteries/other blood vessels; | | 2 |
| (iii) | • distance 13-15 mm;<br>• divide by 10 = 1.3-1.5; | Ecf: max 1 mark if mp1 incorrect but division by 10 of their value is correct | 2<br>Total 12 |

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Describe what red cells and plasma carry, and from where to where

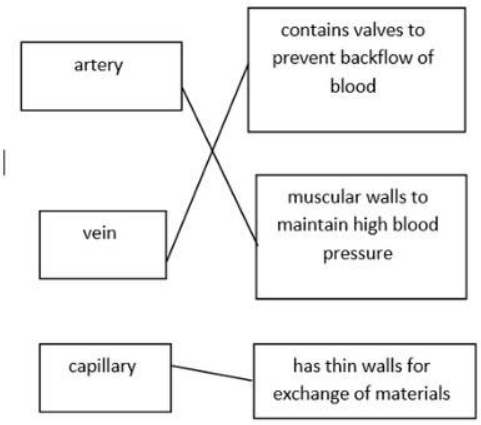
### Answer 3 of 6 · Question 1 part a.i · 3 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Type MS · Describe what red cells and plasma carry, and from where to where

Answer 4 of 6 · Question 2 part a.ii · 2 marks

2024 JUN · Paper 01 (regional R) · find part a.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 2 (a) (i) | X = plasma;<br>Y = red blood cell;<br>Z = white blood cell/phagocyte; | Accept named phagocyte | (3) |
| (a) (ii) | • transports oxygen;<br>• to body/ cells/tissues/organs; | | (2) |
| (a) (iii) | the patient has an infection/leukaemia/allergy/ blood cancer, inflammation; | Accept pathogens in body | (1) |
| (b) |  | Reject more than one line from each blood vessel<br>3 correct=2marks<br>1 or 2 correct= 1mark | (2) |
| (c) (i) | left ventricle; | | (1) |
| (c) (ii) | • into right ventricle;<br>• through pulmonary artery;<br>• to the lungs;<br>• (from the lungs) into the pulmonary vein; | | (4) |

**Total for Question 2 = 13 marks**

Type MS · Describe what red cells and plasma carry, and from where to where

Answer 5 of 6 · Question 5 part b.ii · 2 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b.ii in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 5 (a) | (i) | • axes labelled (1)<br>• suitable scale (1)<br>• independent variable on X axis (1)<br>• correct plots (2)<br>• suitable curve (1) | -1 for each incorrect plot | 6 |
| | (ii) | • mean/average calculated (1)<br>• reliable / reproducible / repeatable (1)<br>• eliminate anomalies (1) | | 2 |
| (b) | (i) | 70 x 100 (1)<br>90<br>78%/77.78/77.8 (1) | correct answer = full marks | 2 |
| | (ii) | • travels in the blood (1)<br>• time to reach/affect nerve (1) | R heart | 2 |
| (c) | | • carries impulses to heart muscle (1)<br>• reduces rate of contraction (1) | | 2 |

Total for question = 14 marks

## Type MS · Describe what red cells and plasma carry, and from where to where

### Answer 5 of 6 · Question 5 part b.ii · 2 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part b.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Describe what red cells and plasma carry, and from where to where

Answer 6 of 6 · Question 5 part c · 3 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

| number | | | |
|---|---|---|---|
| 5 (a) | • addition of figures = 1690;<br>• divide by 10 = 169; | | 2 |
| (b) | makes it faster/decreases reaction time; | | 1 |
| (c) | • caffeine has to be absorbed;<br>• into blood;<br>• transported to nervous system/brain; | Ignore body | 3 |
| (d) | • can't control what is in food/drink;<br>• could contain caffeine/other factors that would affect result/to ensure caffeine is the cause; | Ignore coffee | 2 |
| (e) | 185 ± 5 mm; | | 1<br>Total 9 |

## Type MS · Describe what red cells and plasma carry, and from where to where

### Answer 6 of 6 · Question 5 part c · 3 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|